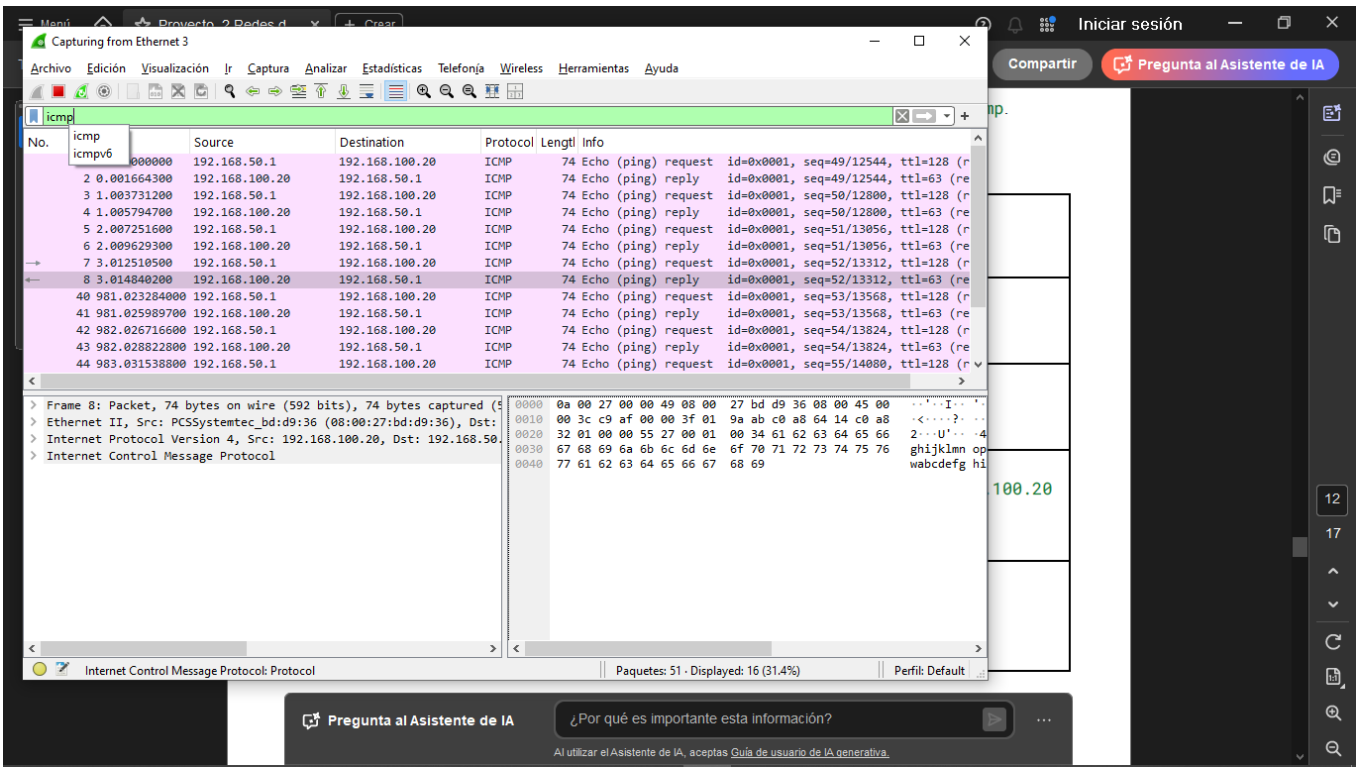
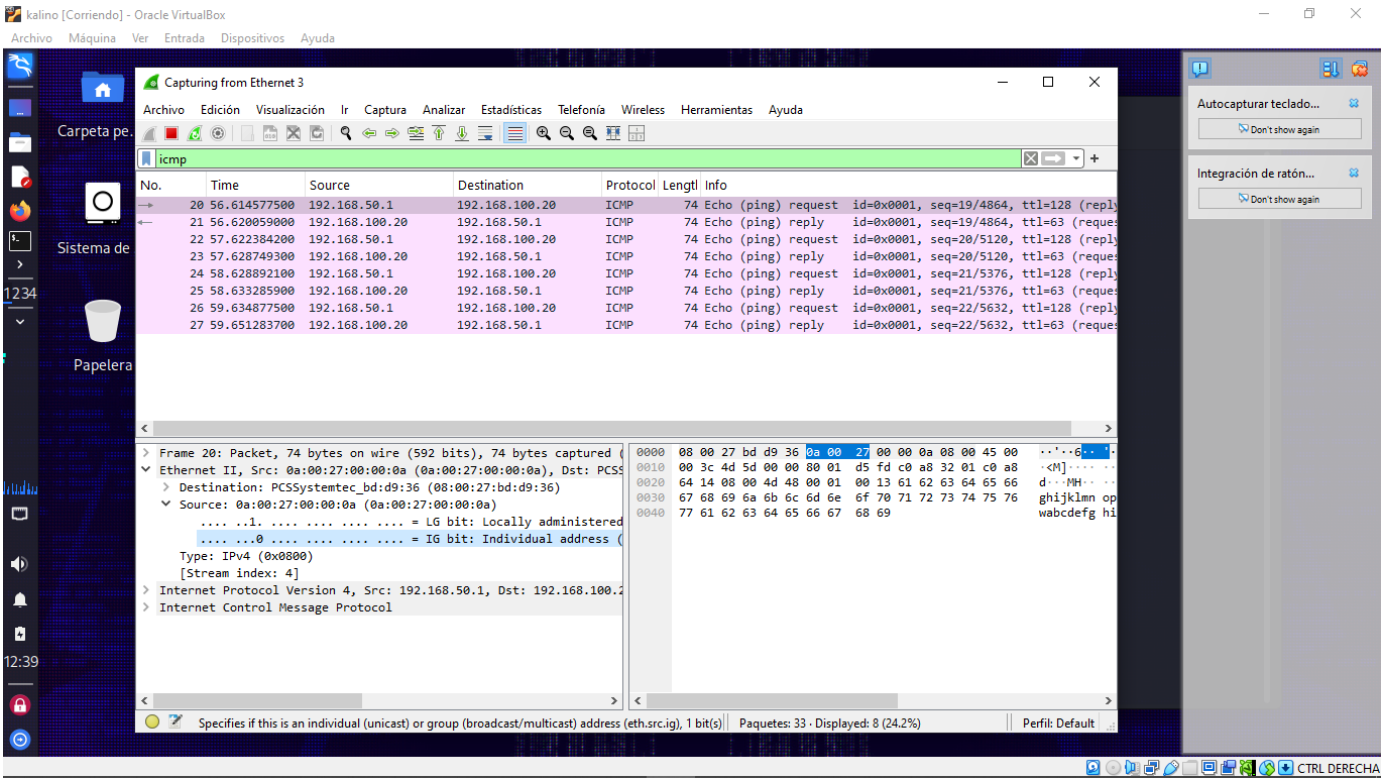
Tabla de Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Mascara Subred	Puerta de Enlace
PC Física (Windows)	VirtualBox Host-Only	192.168.50.1	255.255.255.0	N/A
Router Kali	Adaptador 1 (Host-Only)	192.168.50.10	255.255.255.0	N/A
Router Kali	Adaptador 2 (Interna)	192.168.100.1	255.255.255.0	N/A
Cliente ubuntu	Adaptador 1 (Interna)	192.168.100.20	255.255.255.0	192.168.100.1

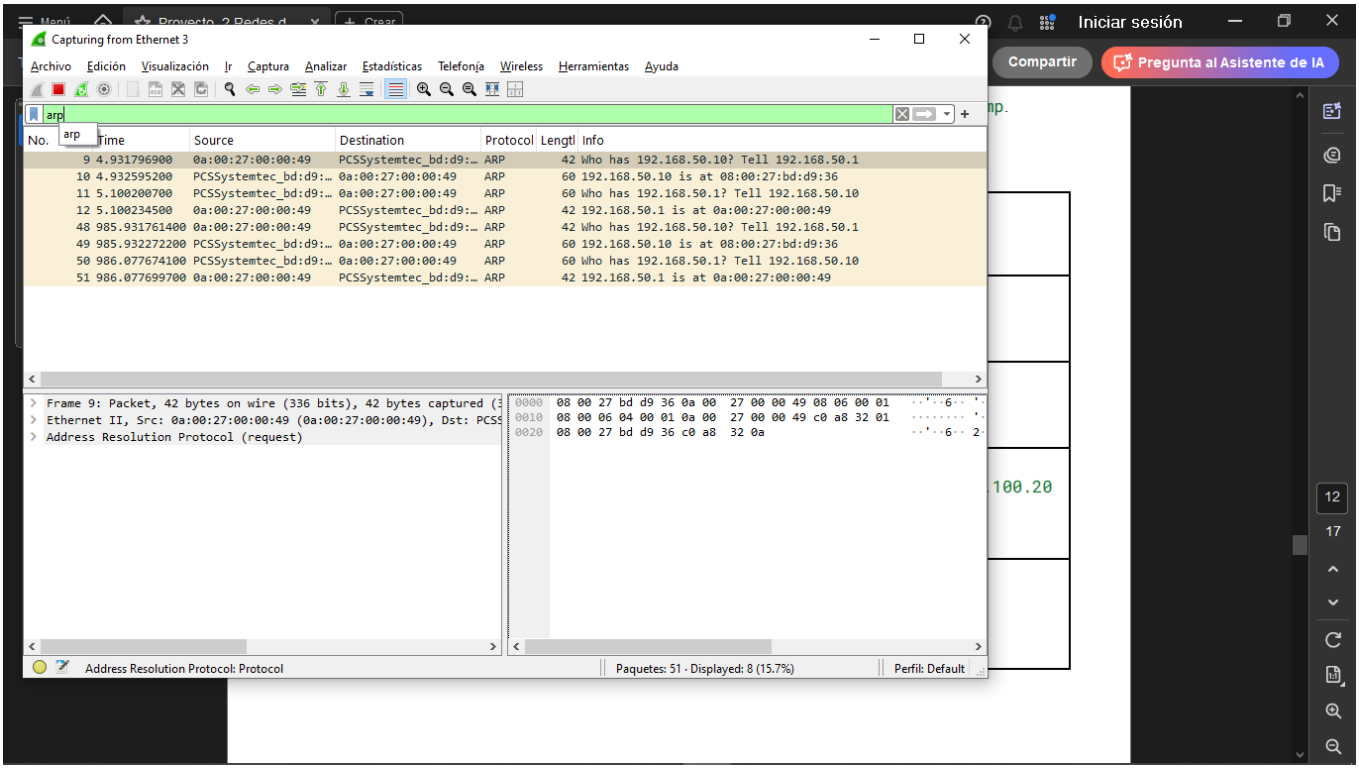
Análisis con Wireshark

ICMP

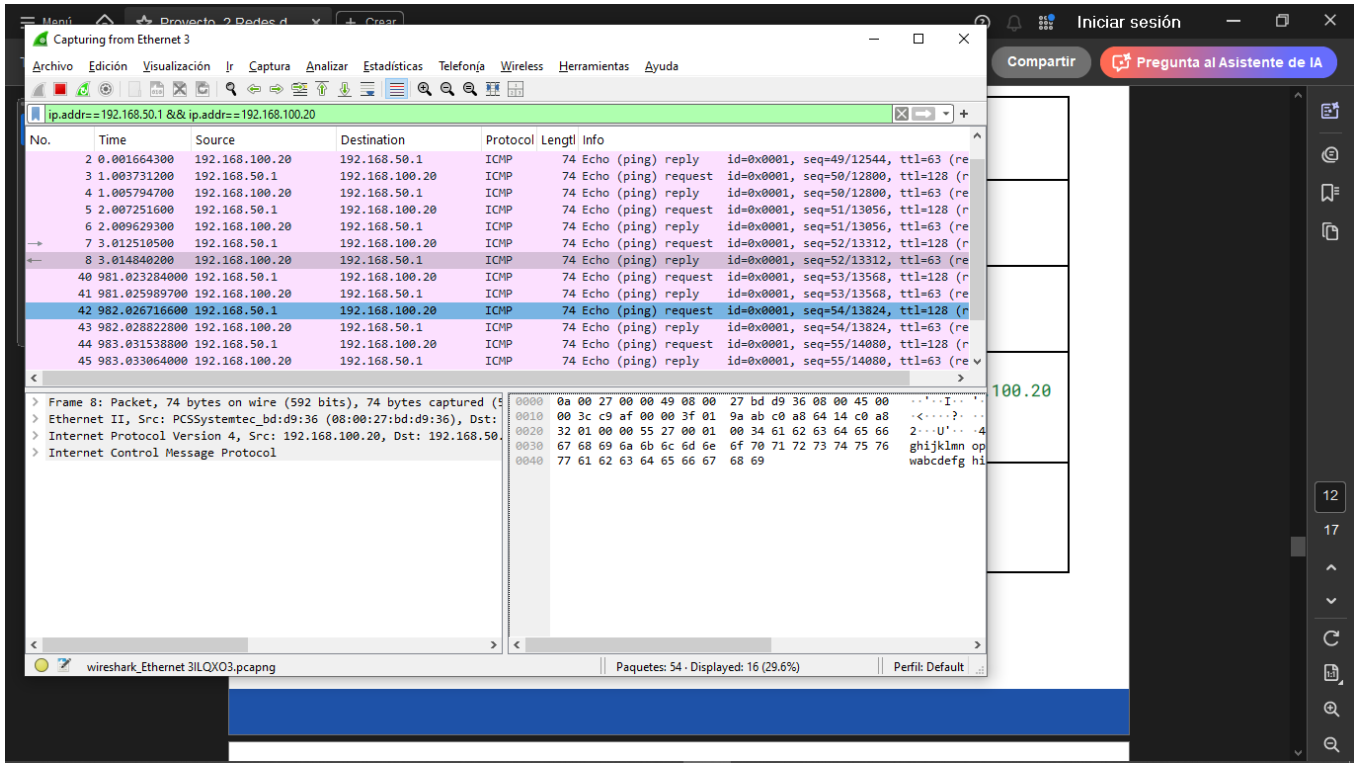



kalino [Corriendo] - Oracle VirtualBox
Archivo Máquina Ver Entrada Dispositivos Ayuda
Capturing from Ethernet 3
Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda
icmp
No. Time Source Destination Protocol Lengtl Info
20 56.614577500 192.168.50.1 192.168.100.20 ICMP 74 Echo (ping) request id=0x0001, seq=19/4864, ttl=128 (reply
21 56.620059000 192.168.100.20 192.168.50.1 ICMP 74 Echo (ping) reply id=0x0001, seq=19/4864, ttl=63 (reques
22 57.622384200 192.168.50.1 192.168.100.20 ICMP 74 Echo (ping) request id=0x0001, seq=20/5120, ttl=128 (reply
23 57.628749300 192.168.100.20 192.168.50.1 ICMP 74 Echo (ping) reply id=0x0001, seq=20/5120, ttl=63 (reques
24 58.628892100 192.168.50.1 192.168.100.20 ICMP 74 Echo (ping) request id=0x0001, seq=21/5376, ttl=128 (reply
25 58.633285900 192.168.100.20 192.168.50.1 ICMP 74 Echo (ping) reply id=0x0001, seq=21/5376, ttl=63 (reques
26 59.634877500 192.168.50.1 192.168.100.20 ICMP 74 Echo (ping) request id=0x0001, seq=22/5632, ttl=128 (reply
27 59.651283700 192.168.100.20 192.168.50.1 ICMP 74 Echo (ping) reply id=0x0001, seq=22/5632, ttl=63 (reques
Frame 20: Packet, 74 bytes on wire (592 bits), 74 bytes captured (
Ethernet II, Src: 0a:00:27:00:00:0a (0a:00:27:00:00:0a), Dst: PCSS
Destination: PCSSystemtec_bd:d9:36 (08:00:27:bd:d9:36)
Source: 0a:00:27:00:00:0a (0a:00:27:00:00:0a)
.... ..1. .... = LG bit: Locally administered
....0 .... = IG bit: Individual address (
Type: IPv4 (0x0800)
[Stream index: 4]
Internet Protocol Version 4, Src: 192.168.50.1, Dst: 192.168.100.2
Internet Control Message Protocol
0000 08 00 27 bd d9 36 0a 00 27 00 00 0a 08 00 45 00
0010 00 3c 4d 5d 00 00 80 01 d5 fd c0 a8 32 01 c0 a8
0020 64 14 08 00 4d 48 00 01 00 13 61 62 63 64 65 66
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
0040 77 61 62 63 64 65 66 67 68 69
Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.src.ig), 1 bit(s)
Paquetes: 33 · Displayed: 8 (24.2%)
Perfil: Default
Autocapturar teclado...
Integración de ratón...
Don't show again
CTRL DERECHA

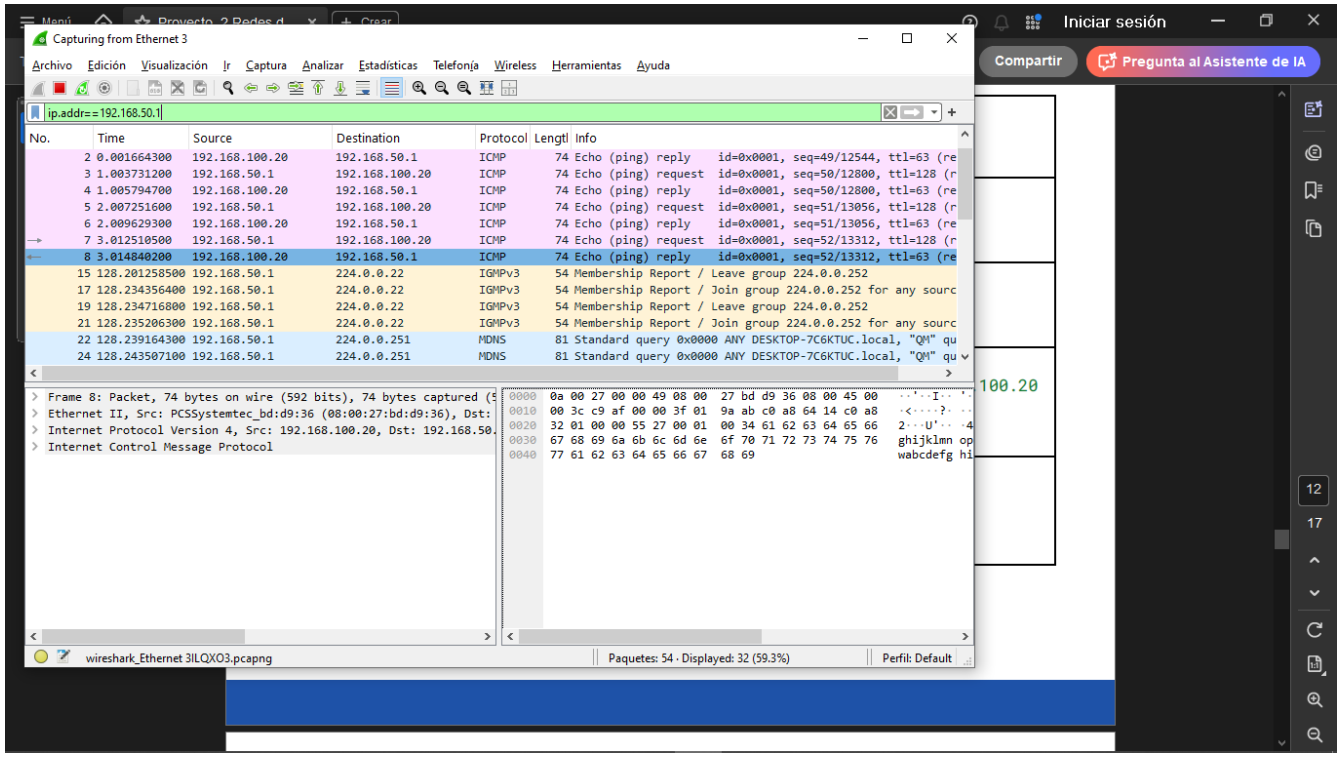
ARP


Iniciar sesión
Compartir
Pregunta al Asistente de IA
Capturing from Ethernet 3
Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda
arp
No. Time Source Destination Protocol Lengtl Info
9 4.931796900 0a:00:27:00:00:49 PCSSystemtec_bd:d9:... ARP 42 Who has 192.168.50.10? Tell 192.168.50.1
10 4.932595200 PCSSystemtec_bd:d9:... 0a:00:27:00:00:49 ARP 60 192.168.50.10 is at 08:00:27:bd:d9:36
11 5.100200700 PCSSystemtec_bd:d9:... 0a:00:27:00:00:49 ARP 60 Who has 192.168.50.1? Tell 192.168.50.10
12 5.100234500 0a:00:27:00:00:49 PCSSystemtec_bd:d9:... ARP 42 192.168.50.1 is at 0a:00:27:00:00:49
48 985.931761400 0a:00:27:00:00:49 PCSSystemtec_bd:d9:... ARP 42 Who has 192.168.50.10? Tell 192.168.50.1
49 985.932272200 PCSSystemtec_bd:d9:... 0a:00:27:00:00:49 ARP 60 192.168.50.10 is at 08:00:27:bd:d9:36
50 986.077674100 PCSSystemtec_bd:d9:... 0a:00:27:00:00:49 ARP 60 Who has 192.168.50.1? Tell 192.168.50.10
51 986.077699700 0a:00:27:00:00:49 PCSSystemtec_bd:d9:... ARP 42 192.168.50.1 is at 0a:00:27:00:00:49
Frame 9: Packet, 42 bytes on wire (336 bits), 42 bytes captured (3
Ethernet II, Src: 0a:00:27:00:00:49 (0a:00:27:00:00:49), Dst: PCSS
Address Resolution Protocol (request)
0000 08 00 27 bd d9 36 0a 00 27 00 00 49 08 06 00 01
0010 08 00 06 04 00 01 0a 00 27 00 00 49 c0 a8 32 01
0020 08 00 27 bd d9 36 c0 a8 32 0a
Address Resolution Protocol: Protocol
Paquetes: 51 · Displayed: 8 (15.7%)
Perfil: Default
100.20

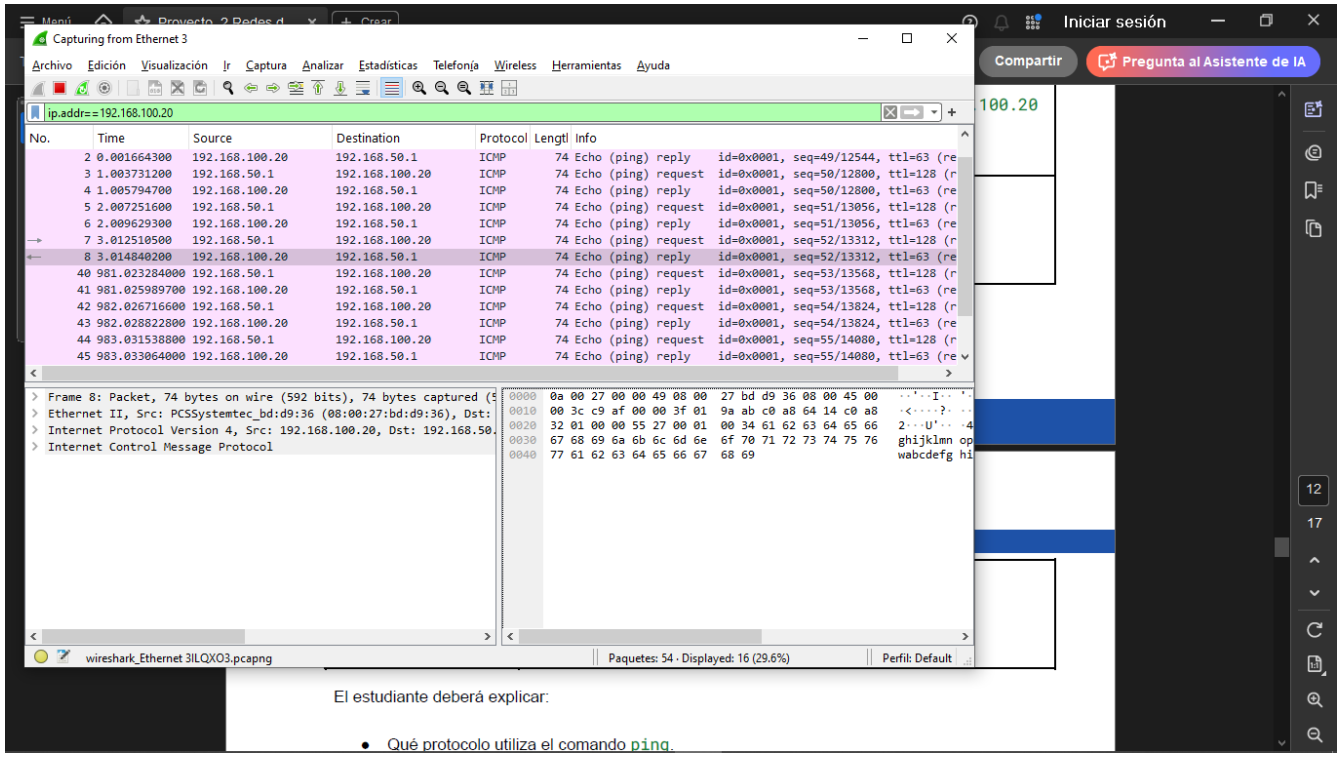
ip.addr == 192.168.50.1 && ip.addr == 192.168.100.20



ip.addr == 192.168.50.1



ip.addr== 192.168.100.20



Preguntas de Análisis

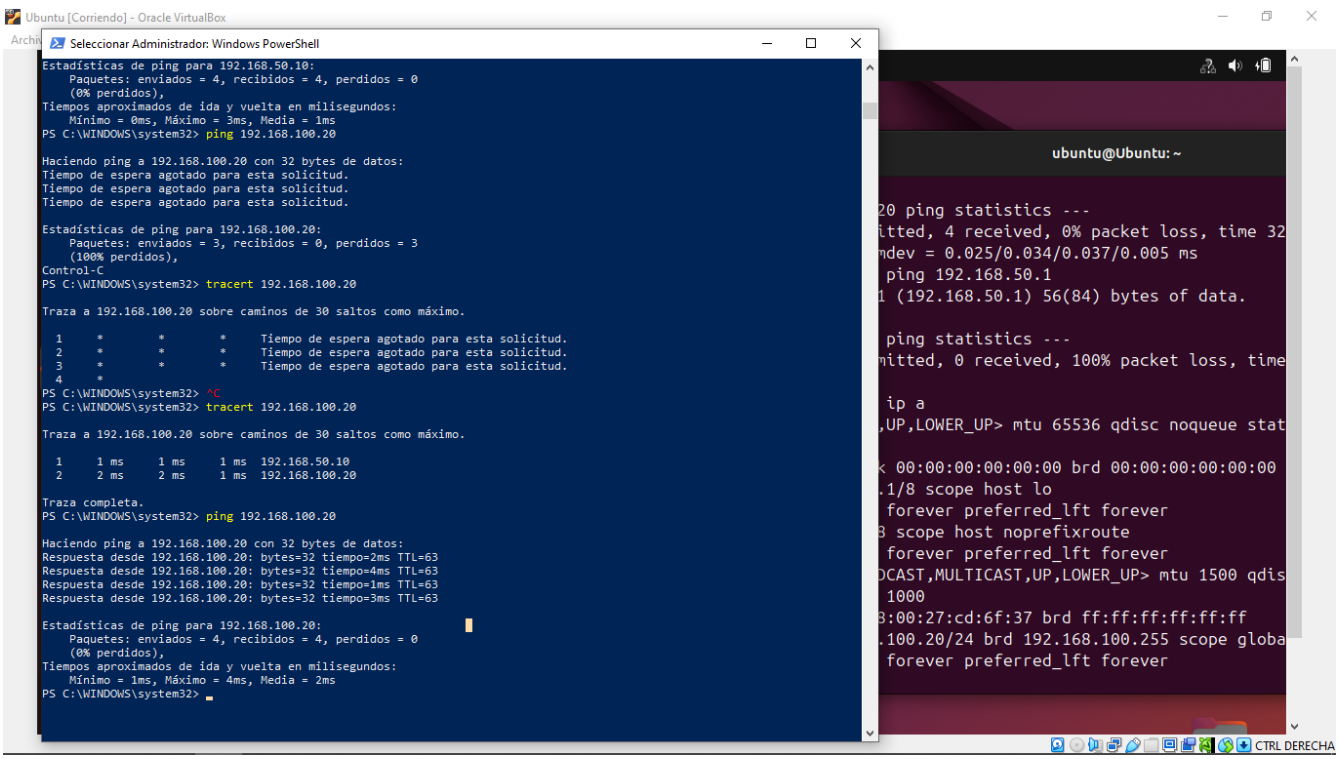
¿Qué protocolo utiliza el comando ping? Utiliza el protocolo **ICMP** (Internet Control Message Protocol).

¿Para qué se utiliza ARP? Se utiliza para asociar una dirección IP conocida con una dirección MAC física en la red local.

¿Qué dirección MAC se utiliza como destino cuando el tráfico va hacia otra red? Se utiliza la **dirección MAC de la interfaz del Router (Gateway)**. En tu caso, cuando Windows le manda un ping a Ubuntu, la MAC de destino en la trama será la de Kali (192.168.50.10), no la de Ubuntu.

¿Qué ocurre con el TTL cuando el paquete pasa por el router? El valor del **TTL (Time To Live) disminuye en 1** cada vez que atraviesa un router (salto de capa 3). Por eso viste que el ping a Ubuntu te marcaba TTL 63 en lugar de 64.

Prueba de verificación



Comando de permitir paquetes

sudo sysctl -p